Functional Requirements

Functional Requirement	Value and Unit	Type
Material To Engrave	Acrylic	N/A
Variation in Depth of Cut (Z-repeatability)	.002"	Maximum
In-Plane variation (XY-Repeatability)	.004"	Maximum
Depth of Cut	.005"	Maximum
Feed Rate	12 in/min	Maximum

Theoretical Motion Resolution for all motion Axes

a) The following equation is used to calculate the finest resolution of motion that can be obtained with a single microstep

$$dist = \frac{1}{(\ stepper\ motor\ steps)(microsteps)}(1\ \mu step)(screw\ lead)$$
$$= \frac{1}{\left(200\frac{steps}{rev}\right)\left(16\frac{\mu steps}{step}\right)}(1\ \mu step)\left(4\frac{mm}{rev}\right) = \ .0013\ mm = 1.3\ \mu m$$

b) As each axis is using the same stepper motor and the same lead screw and we are neglecting other factors (applied loads, friction, etc), I believe it is reasonable to assume that this minimum value will be the same for all axes.

c) The current dial indicator we are given can measure tick marcs of 0.005" (0.127 mm) and the human eye can probably estimate about a fourth of a tick mark 0.00125" (0.0318 mm). Just using the tick marks, the dial indicator can only read .100 times the minimum value and even using partial tick marks, the dial indicator can only read 25 times the minimum value, meaning that the dial indicator would not be able to reliably read a change in translational distance of the minimum value.

d) This idealized motion being many times smaller than both the previously measured error sources and the resolution of the dial indicator leads to the assumption that the stepper motor resolution is not the limiting factor for this machine's precision.

Friction in the leadscrew drive and elastic backlash

a) Torque threshold calculation

b) This equation assumes the applied load is that of the cutting force calculated previously of 10N, the coefficient of friction *f* = 0.2, a mean diameter d_m7mm, the lead *l*=4mm, an internet search said that a common acme leadscrew the thread angle α was 15 deg, so that is a reasonable assumption to begin with. This equation also neglects any geometric backlash due to the screw and nut not physically touching. By using the cutting load as the applied load, the

$$T_R = F\frac{d_m}{2}\left(\frac{1+\pi f\ d_m \sec(\alpha)}{\pi d_m - fl\sec(\alpha)}\right) = \frac{10N\ (.007m)}{2}\left(\frac{1+\pi(.2)(.007m)\sec(15\ deg)}{\pi(.007m)-(.2)\left(.004\frac{m}{rev}\right)\sec(15\ deg)}\right) = 1.66\ Nm$$

c) In this context, vertically means that the load is acting in line with the axis of the lead screw and the direction of motion is in line with the gravity vector of the applied force. In our case, the applied load is not necessarily in line with the direction of the axis of the leadscrew. This equation does not include additional effects of bending stresses on the leadscrew and leadscrew misalignment and stiffness effects. If the leadscrew is bent due to an applied load, it will have an increased contact force between that of the nut and the screw and may take more input torque to raise/lower the load. Depending on the applied load, for the purposes of this application, it is reasonable to include it as a lower bound of torque required when the material is engaged with the tool and an upper bound of torque if the router is not actively cutting.. Although we are using the worst-case cutting loads, we are also not including the effects of bending stress loads.

d) The torsional leadscrew stiffness is calculated as follows:

$$k_\theta = \frac{\tau}{\theta} = \frac{GJ}{L}\left(\frac{2\pi\, rad}{360\text{deg}}\right) = \frac{G\left(\pi\frac{r_m^4}{2}\right)}{(.33m)}\left(\frac{2\pi\, rad}{360\text{deg}}\right) = \frac{(77GPa)\left(\pi\frac{r_m^4}{2}\right)}{(.33m)}\left(\frac{2\pi\, rad}{360\text{deg}}\right) = .96\frac{Nm}{deg}$$

Where G is the modulus of rigidity for steel, J is the polar moment of inertia, rm is the mean radius of the leadscrew, L is the length of the leadscrew and coupler in the Y-axis. The worst case configuration is simply if the entire leadscrew is in torsion. Although this is somewhat conservative because the leadscrew nut on either carriage is never actually at the exact end of the leadscrew, it is a good starting point to see if further analysis is required.

e) The stepper motor rotation needed to break friction of the leadscrew nut can be calculated as follows

$$\tau_R = k_\theta \theta_f \rightarrow \theta_f = \frac{\tau_R}{k_{theta}} = \frac{1.66\, Nm}{.96\frac{Nm}{deg}} = 1.72\, deg \rightarrow (1.72 \text{ degrees})\left(\frac{4mm}{360degrees}\right) = .0192\, mm = 19.2\, \mu m$$

f) In this case the leadscrew elastic backlash is larger than the resolution of then that of the stepper motor microstepping, but should be achievable without overshooting very much and ‘sling-shotting’ the carriage on the leadscrew.

g) In this case, the backlash will affect the machine repeatability when considering both the backlash of the X and Y axes (assuming the z axis does not really vary during the engraving process). The overall functional requirement is that the repeatability should be less than $100\, \mu m$, and the combination of the leadscrew backlash takes into account nearly half of that amount. This in combination with the structure stiffness means that that the repeatability requirement is not able to be met. A stiffer (larger diameter) leadscrew with less friction may be able to reduce the backlash error.

Friction in the leadscrew drive and stick-slip motion

a) The calculation for the torque threshold using the dynamic coefficient of friction is the same as the previous calculation, although using a different value for the coefficient of friction, *f*.

$$T_R = F\frac{d_m}{2}\left(\frac{1+\pi f\, d_m \sec(\alpha)}{\pi d_m - fl\sec(\alpha)}\right) = \frac{10N\,(.007m)}{2}\left(\frac{1+\pi(.15)(.007m)\sec(15\, deg)}{\pi(.007m)-(.15)\left(.004\frac{m}{rev}\right)\sec(15\, deg)}\right) = 1.64\, Nm$$

b) In this case, there is not a meaningful difference between the static and dynamic required torque thresholds. According to this calculation, it will take 0.02Nm less torque to keep moving the leadscrew after the static friction is broken.
Using the prior calculation to calculate rotation of the leadscrew, once the load stops moving the leadscrew rotates:

$$\theta_d = \frac{\tau_{diff}}{k_{theta}} = \frac{1.64\, Nm}{.96\frac{Nm}{deg}} = 1.70\, degrees$$

c) The “stick slip” amount is calculated as follows

$$stickSlip = 1.72\, degrees - 1.70\, degrees = 0.2 \text{ deg} = 0.0022\, mm = 2.2\mu m$$

d) This value is nearly double that of the microstep minimum step value, meaning that the stepper motor will be off by two microsteps every time the motor stops moving. For a very long cut this may add up, but for a short cut the difference may not be noticeable as it is much less than the overall system repeatability requirement and much less than the load induced errors (from the previous assessment) and the backlash estimated.

Repeatability of X-Axis Leadscrew using Dial Indicator

a) I chose to characterize the x-stage motion. I began by measuring the repeatability with the carriage near the middle of the X-stage and I clamped the dial indicator to the Y-carriage such that it was preloaded against the smooth plastic piece holding the spindle. I began by moving the stage back and forth in 0.2mm increments, noting the initial, middle and final positions of the dial indicator. I repeated this experiment five times. I then repeated this process using 2mm increments and 20mm increments. I then moved the dial indicator and X-carriage to the ‘far’ side of the linear stage, away from the motor, and then on the “near” side of the linear stage, towards the motor. The following equation was used to estimate the repeatability of the X-axis

$$\mu_{true} = \mu_{meas} \pm t_{.025}\frac{\sigma}{\sqrt{n}}$$

Where μ_{meas} is the mean of the measured data, σ is the standard deviation of the data points, *n* is the number of data points per trial and $t_{.025}$ is the t-factor used for scaling the data for computing uncertainty in small data sets.

	Carriage Position Repeatability One-Way Motion Measured at Midpoint		
Step Size	**Near Position**	**Mid Position**	**Far Position**
0.2mm	$.19\ \pm 0\ mm$	$.178 \pm 0\ mm$	$.175\ \pm 0\ mm$
2mm	$1.99\ \pm 0\ mm$	$1.97\ \pm 0\ mm$	$1.97\ \pm 0.01\ mm$
20mm	$19.92 \pm 0\ mm$	$19.90\ \pm 02\ mm$	$19.88\ \pm 0.01\ mm$

	Carriage Position Repeatability (Return to Original Position) Measured at return to Zero		
Step Size	**Near Position**	**Mid Position**	**Far Position**
0.2mm	$0\ \pm 0\ \mu m$	$2.5 \pm 3.6\ \mu m$	$0\ \pm 0\ \mu m$
2mm	$0\ \pm 0\ \mu m$	$2.5\ \pm 5.8\ \mu m$	$2.7\ \pm 4.5\ \mu m$
20mm	$1.9\ \pm 1.8\ \mu m$	$-.6\ \pm 3.6\ \mu m$	$-1.3\ \pm 18\ \mu m$

b) Based on my predictions, the stepper motor rotation required to break free of the frictional torque results in a net loss of motion of 0.019mm of motion of the leadscrew. The measured values indicated that this approximation is off by approximately one order of magnitude as many of the measurements regardless of position on the leadscrew being off approximately 0.1mm from the commanded target. This could potentially be due to additional frictional effects that aren’t accounted for in the torsional analysis, such as that of the seals on the linear bearings on the rails.

c) The general trend of the data shows that there is a difference in repeatability between 0.2mm stroke length and the 20mm stroke length, but they are still on the same order of magnitude. Although the repeatability is slightly higher for the longer stroke lengths, I don’t necessarily see a meaningful trend due to the resolution of the dial indicator.

d) In general, I noticed the precision decreasing in accuracy the further along the length of the leadscrew the measurement was taken. When the carriage was close to the leadscrew, the measurements varied from 0.01-0.1 mm from ideal, in the middle the measurements varied from 0.022-0.1mm from ideal, and in the far position the measurements varied from 0.025mm to 0.12mm from ideal. The largest variations occurred when the stroke length was larger.

e) Overall, I did notice a drift in measurements, particularly as the stroke length increased, regardless of position along the shaft. Although in this case the drift was noticeably worst and had larger variance in the far position with the largest stroke length.

Additional Error Investigation

a) I chose to investigate the machine drift over many repeated motions
b) Based on my previous leadscrew calculations, we lose approximately 19.2 μm every time the x-carriage is stopped and re-started. Over the course of 50 motions, I would expect to lose 0.96mm over the course of a very long run.
c) This error would manifest by warping the desired shape of the engraving over the course of the cut, particularly if there are a lot of stops and starts of one of the axes. This can be amplified further if you take into account a similar error in the Y-axis. This error could be measured by engraving known geometric shapes and then measuring the deviation in those shapes, such as the parallel-ness of diagonal lines or a square zig-zag pattern.
d) . To investigate this, I started by setting up the dial indicator to measure the X-axis carriage starting from the end of the leadscrew closest to the motor. I then moved the X carriage back and forth 50 times using a feed rate of 500 and a stroke length of 200mm and took a measurement at the end of the 100 movements.
e) After the 50 back and forth trials, the dial indicator read a displacement of 0.127mm, indicating the amount of translational motion lost due to friction in the linear stage assembly.
f) The measured value is actually remarkably close to the predicted 0.96mm of expected loss. It is reasonable that the measured error is larger than the predicted in this case because the original frictional loss calculation does not account for additional loads on the system, such as the friction in the linear bearings or the additional forces due to rail/screw misalignment. In this case, although the error is outside the functional requirement for repeatability, it is not the largest source of error considering the structural loading errors.

System Improvements

a) Based on all of the prior analysis from the original structural assessment and the repeatability assessment, I believe the largest contributor to the error of the system comes from the compliance in the structure, particularly the plastic X-Z carriage and the linear stage compliance. There are many potential changes that could be made to the system to eliminate sources of error, such as making the linear rail bearing blocks out of a single piece of aluminum to promote less rail misalignment. Another change that could be made would be to modify the way the leadscrew is attached to the carriages such that it uses a flexure to avoid overconstraint forces. Another change that could be made would be to swap the set screws that attach to the leadscrew for a proper shaft coupling attachment, such as how we mount our lathe spindle shaft. Another change that I could make would be to change the mounting of the leadscrews such that the applied loads that the screw sees are directed through the machine structure rather than into the stepper motor.
b) All in all, there are a lot of small changes that could be made to improve the machine, but for the sake of time I would like to discuss one small change that could be made that would greatly improve the machine is the X-axis rail and x-axis carriage. When I apply a load to the collet, I can visibly see the torsion effect acting through the plastic of the x-carriage and the movement of the rails of the x-axis. Based on my HTM analysis, I believe that by simply modifying the diameter of the rods we can increase the stiffness greatly.

c) Below are two images, the first on the left is the HTM output summary of the errors induced using the standard design, and the second is the HTM summary of the errors induced by simply changing the diameter of the rails from 10mm to 15mm. As you can see, the overall output stiffness in all three axes are within the functional requirement tolerance specified in my structural analysis report.

10mm guide rails (left)

SumXNom	0
SumYNom	-2.9E-05
SumZNom	-1.4E-14

SumXErr	-0.00858
SumYErr	0.92521
SumZErr	-0.19547

Total Error				
dx	-0.00858	mm	-8.58106	um
dy	0.925239	mm	925.239	um
dz	-0.19547	mm	-195.469	um

Abbe Errors				
dx	-0.0628	mm	-62.7501	um
dy	0.8775	mm	877.5315	um
dz	0.2345	mm	234.4623	um

Loading Errors				
dx	0.0819	mm	81.87182	um
dy	0.0713	mm	71.25994	um
dz	0.0389	mm	38.86151	um

Stiffness	
kx	122.1421
ky	140.3313
kz	257.324

15mm guide rails (right)

SumXNom	0
SumYNom	-2.9E-05
SumZNom	-1.4E-14

SumXErr	-0.00858
SumYErr	0.24171
SumZErr	-0.04408

Total Error				
dx	-0.00858	mm	-8.58106	um
dy	0.241738	mm	241.7383	um
dz	-0.04408	mm	-44.082	um

Abbe Errors				
dx	-0.0175	mm	-17.524	um
dy	0.2431	mm	243.0659	um
dz	0.0686	mm	68.60251	um

Loading Errors				
dx	0.0251	mm	25.1011	um
dy	0.0222	mm	22.21003	um
dz	0.0105	mm	10.47615	um

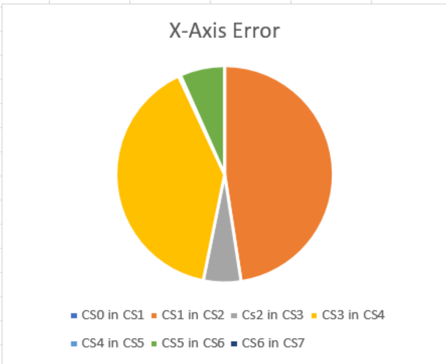
Stiffness	
kx	398.389
ky	450.2471
kz	954.5493

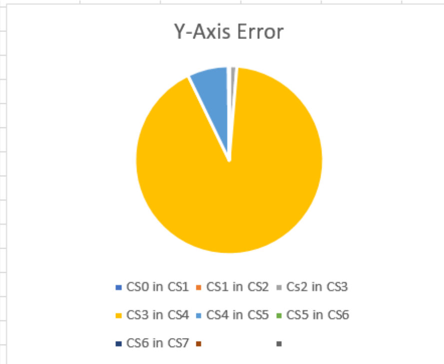
Error Summary using 10mm guide rails (left) and 15mm guide rails(right)

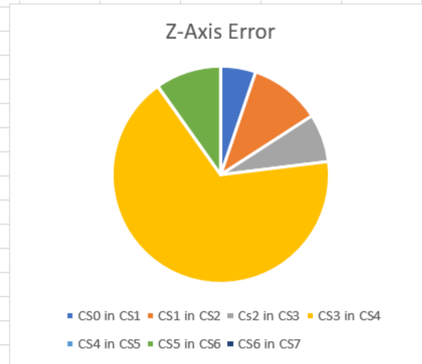
d) A Tight-Tolerance Ultra-Machinable 12L14 Carbon Steel Rods Mcmaster Carr (https://www.mcmaster.com/5544t312) that is 3 feet in length and 10mm in diameter is $10.07. The same bar with 15mm diameter is $18.87. Therefore, it would be a $32 increase in cost to upgrade the X and Y rails to a larger diameter. Considering the entire router is sold for $200, this price increase may be too much, however it may be doable considering that the rails may actually be much cheaper wholesale.

HTCompliance Model

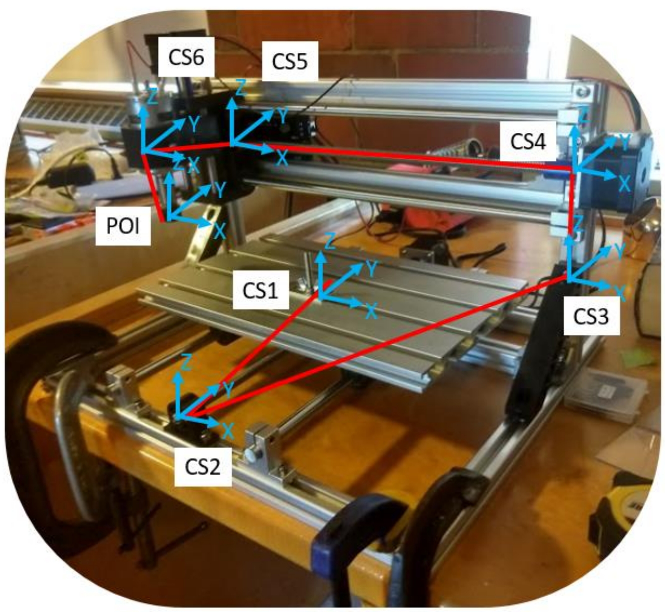
a) Summary of Findings







Here is a summary of the errors modeled in my router. The primary source of error in all three systems seems to be around coordinate system number 4, which is attached to the base of the X-axis leadscrew. In this case, this intersection gets hit with a double whammy from the cantilever beam stiffness of the 8020 gantry and the abbe error from the compliant rails.



Ranking (Most to least)	X-Axis Error Sources	Y-Axis Error Sources	Z-Axis Error Sources
1	Plastic Z-Axis Cantilever Loading (bearings + plastic)	Plastic Z-Axis Cantilever Loading (bearings + plastic)	Plastic Z-Axis Cantilever Loading (bearings + plastic)
2	X-Axis Leadscrew backlash	X-Axis Torsional stiffness, Abbe error	X-Axis rail stiffness
3	X-axis torsional stiffness	8020 cantilever stiffness	Y-axis rail stiffness
4	Y-axis linear rail x-stiffness	Y-axis leadscrew repeatability	Z-axis leadscrew backlask
5	Z-axis linear rail x-stiffness	Bearing Stiffness	8020 cantilever axial stiffness + abbe error
6	Bearing stiffness		z-axis leadscrew stiffness
7			Bearing Stiffness

Simple first order analysis allowed me to get an initial feel for what sources of error were the most critical to analyze in the htm. Among those that could be easily eliminated were the axial stiffness of the Y-carriage plate and the axial stiffness of the 8020 gantry. Further, in my recent experience with the spindle design, the bearings were much more stiff than the spindle shaft, therefore I used that logic to put the bearing stiffness at the low end of my error list.

By comparing the stiffness summary included in my structural assessment with the values of stiffness modeled in my HTM, in many respects, my modeled stiffness is much larger than what is measured. For example, my modeled stiffness for the gantry system assumes a perfectly cantilevered set of beams while in practice I neglected to inclied the stiffness of the bolted connections to the 8020. In this case, my measured stiffness was [kx ky kz]=[400 290 inf] N/mm and my modeled stiffness is [kx ky kz]=[3111 3111 33E4] N/mm. However, despite these variations in stiffness, the abbe errors still play a very large part in the overall estimated error of the system. The stiffness that I modeled for the entire system is [kx ky kz]=[122, 140 257] N/mm and the measured stiffness is [kx ky kz]=[76 277 39] N/mm. Overall these values are on the same order of magnitude and are relatively close considering I may have been overly conservative with some estimates (cantilever beams, etc.) and under-conservative with others (neglecting bearing stiffness, etc). Overall I feel that my current error analysis, although admittedly incomplete, does a relatively accurate job of predicting the possible range of stiffness and accuracy values one might expect from using this machine. Below is a summary of my HTM modeling errors.

One thing that I did not have time to do but would have been valuable insight is to do the famous "Slocum laser test" to find the running parallelism errors for each of the linear stages. As this router is assembled by hand without any guiding markings or features and the components are all separate, it would be difficult to align them perfectly. Even by using the straightness of the rails to align the carriage bearings and then using the bearings to align the pillow blocks, there is still significant room for human error that can manifest even if the table slides smoothly.

On a side note, I feel like my HTM spreadsheet is pretty close to being actually good, so any advice or suggestions you have would be greatly appreciated. I would love to be able to teach other students how to use HTMs properly using mine as an example.

Actual Coordinates of POI in CS1 due to loads and errors

	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords	
x	0.00000000	0.00000000	0.00000001	0.00000000	0.00000001	0.00000000	mm
y	-22.52860000	0.00000000	0.00000001	0.00000800	0.00000801	-22.52860801	mm
z	0.00000000	0.00000000	0.00000001	-0.01898134	0.01898135	0.00000000	mm

Actual Coordinates of CS1 in CS2 due to loads and errors

	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords	
x	0.00000000	0.00000000	0.07745553	0.00000000	0.07745553	0.00000000	mm
y	165.50000000	0.00000000	0.00038996	0.00000000	0.00038996	165.50038996	mm
z	78.25000000	0.00000000	0.03872777	0.00000000	0.03872777	78.28872777	mm

Actual Coordinates of CS2 in CS3 due to loads and errors

0.00000000	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords	
x	-17.15000000	0.00000000	0.00000001	0.00910389	0.00910390	-17.15910390	mm
y	-234.50000000	0.00000000	0.00000001	0.01217799	0.01217800	-234.51217800	mm
z	-112.45000000	0.00000000	0.00000001	-0.02612004	0.02612005	-112.47612005	mm

Actual Coordinates of CS3 in CS4 due to loads and errors

	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords	
x	0.00000000	0.00000000	0.00321390	-0.06170451	0.06491840	0.00000000	mm
y	20.50000000	0.00000000	0.00321390	0.86600681	0.86922071	21.36922071	mm
z	-74.45000000	0.00000000	0.00002971	0.24391908	0.24394879	-74.69394879	mm

Actual Coordinates of CS4 in CS5 due to loads and errors

	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords	
x	17.15000000	0.00000000	0.00046796	-0.00005488	0.00052284	17.15052284	mm
y	0.52857143	0.00000000	0.06692158	-0.00020143	0.06712301	0.59569444	mm
z	0.55000000	0.00000000	0.00000000	0.00018733	0.00018733	0.55018733	mm

Actual Coordinates of CS5 in CS6 due to loads and errors

	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords	
x	0.00000000	0.00000000	0.00073439	-0.01009460	0.01082900	0.00000000	mm
y	97.00000000	0.00000000	0.00073439	-0.00020843	0.00094282	97.00094282	mm
z	0.55000000	0.00000000	0.00010399	0.03551917	0.03562316	0.58562316	mm

Actual Coordinates of CS6 in CS7 due to loads and errors

	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords	
x	0.00000000	0.00000000	0.00000001	0.00000000	0.00000001	0.00000000	mm
y	-26.50000000	0.00000000	0.00000007	-0.00025140	0.00025147	-26.50025147	mm
z	107.55000000	0.00000000	0.00000001	-0.00006194	0.00006195	107.55006195	mm

Actual Coordinates of CS7 in CS8 due to loads and errors

	Nominal	Random Errs	Loading Errors	Abbe Errors	Sum Errors	Coords	
x	0.00000000	0.00000000	0.00000001	0.00000000	0.00000001	0.00000000	mm
y	0.00000000	0.00000000	0.00000001	0.00000000	0.00000001	0.00000000	mm
z	0.00000000	0.00000000	0.00000001	0.00000000	0.00000001	0.00000000	mm